Game Title: Whisker Dash

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1. Summary of the Game Concept

Whisker Dash is a fast-paced 2D arcade survival game where the player controls a cat moving through a hidden city alley environment filled with obstacles (Trash cans) and enemies (Dogs and Rats). The main objective of the game is to survive for as long as possible while defeating enemy dogs and rats that appear on the map. The cat attacks by launching scratch-like projectiles in the direction it is facing, allowing the player to fight back against incoming threats.

To stay alive, the player must avoid taking too much damage from enemies and obstacles while collecting health packs in the form of cat food to restore lost health. Each defeated enemy increases the player's score, and the goal is to reach a score of 100 to win the game. Player speed gets increased over time.

The game is designed to provide a simple but engaging arcade-style experience that combines movement, aiming, timing, and survival mechanics. It focuses on quick reflexes, player awareness, strategic positioning, and managing health while dealing with different enemy behaviors in a confined environment.

2. Description of Core Features

Player Movement

The player movement is implemented using keyboard input (WASD and arrow keys) together with Rigidbody2D velocity. This allows the cat character to move smoothly in four directions.

Projectile Attack System

The shooting system is implemented by instantiating projectile objects from the player's position. The projectile travels in the direction the player is facing and is destroyed after collision or after a short time.

Dog Enemy Patrol

The dog enemy is implemented as a patrol-based enemy that moves continuously between two assigned points (Point A and Point B). This movement is handled using a movement script. The dog damages the player on contact and requires three projectile hits to be defeated.

Rat Enemy Spawning

The rat enemy is implemented using rat spawn zones placed behind trash can obstacles. This creates the effect of rats appearing from hidden alley areas when the player moves near those spawn points.

Rat Movement Behavior

After spawning, rats move toward the player using a zig-zag movement pattern. They follow the player for a limited time and are automatically destroyed after a short duration if not defeated. Rats require only one projectile hit to be eliminated.

Obstacle Interaction

Walls and trash cans are implemented using colliders. Walls act as solid barriers that block movement, while trash cans function as harmful obstacles that reduce player health when touched.

Health System

The player health system is implemented using a numerical health value linked to a UI slider. Health decreases when the player is damaged and increases when health packs are collected. Health bar decreases and increases in a gradient of colors.

Health Pack Spawn System

Health packs are implemented with a spawning system that makes them appear at selected spawn locations, remain for a few seconds, disappear, and then reappear again later. This adds challenge and timing to health recovery.

Damage Feedback Effects

When the player takes damage, a red screen flash and camera shake effect are triggered. These visual effects are used to make enemy and obstacle collisions feel more impactful.

Healing Feedback Effect

When the player collects a health pack, a green screen flash is triggered. This provides clear visual feedback that health has been restored.

Dynamic Health Bar

The health bar includes a smooth color gradient effect. Instead of changing only at fixed levels, it gradually transitions from green to yellow, orange, and red based on the player's current health amount.

Scoring System

The scoring system is implemented so that points are awarded whenever enemies are defeated. Rats give smaller points, while dogs provide a higher score reward.

Win and Game Over System

The game state is controlled using a GameManager script. The player wins when the score reaches 100 and loses when health reaches zero. Separate UI panels are displayed for the win and game over screens.

3. Screenshots of Gameplay

Figure 1 : Game start

Figure 2: Player navigating the environment

Figure 3: Player combat with enemy dog using projectile

Figure 4: Screen flash green when player gets a health pack (cat food)

Figure 5: Screen shake and flash red when player touch an enemy

Figure 6: Game win screen at score 100

Figure 7: Enemy (mouse) following player

Figure 8: Health decrease in color gradient

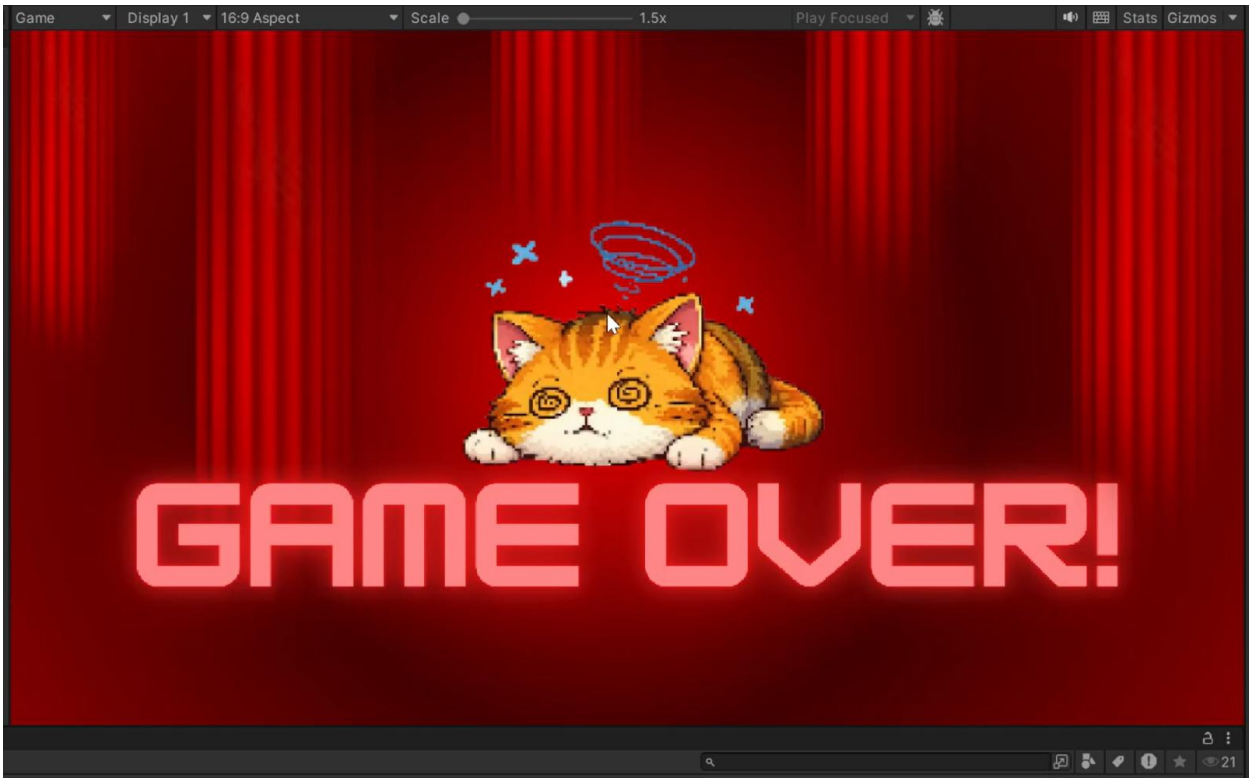

Figure 9: Game over screen when health is 0

4. Control Guide

Movement:

W or Up Arrow - Move Up
S or Down Arrow - Move Down
A or Left Arrow - Move Left
D or Right Arrow - Move Right

Attack:

Left Mouse Click - Shoot projectile

Objective:

Defeat enemies and survive to reach score 100.

5. Description of the creative features

Several creative features were added to improve the overall gameplay experience and make the game more visually and interactively engaging.

- One of the main creative features is the dynamic health bar color system. Instead of changing the health bar color only at fixed points, the health bar smoothly transitions through a gradient of colors based on the player's current health. It changes from green when health is high, to yellow and orange at medium levels, and finally to red when health is low. This provides continuous visual feedback and helps the player quickly understand their current health condition.
- Another creative feature is the damage feedback effect when the player is attacked by enemies. When the cat takes damage from enemy dogs, rats, or obstacles, a red screen flash appears along with a screen shake effect. This combination gives immediate visual feedback to the player and makes enemy collisions feel more impactful and intense.
- A healing feedback feature was also implemented for health recovery. When the player collects a health pack in the form of cat food, the screen briefly flashes green. This helps clearly communicate that health has been restored and creates a satisfying reward effect for collecting healing items.
- One of the enemies start to follow the player for few seconds

- In addition to these, several sound effects were added to improve immersion and make gameplay feel more responsive. Sounds were included for actions such as shooting, taking damage, healing, increasing score, winning, and losing the game.

6. Credits for external Assets/ Tools

Audio: Background music and sound effects from Pixabay (https://pixabay.com)

Graphics: Custom pixel art created using AI tools and image editing software and Canva

7. Video link

SE4031 2D Game IT22179258 Gameplay.mp4

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8. Zipped Unity Project folder

SE4031_2D Game_IT22179258_Project_folder.zip